

Investigator Brochure: RenalGuard-5

1. PHARMACOKINETICS & MECHANISM

RenalGuard-5 is a selective endothelin receptor antagonist designed to slow the progression of chronic kidney disease through unique dual-pathway renoprotection.

Mechanism 1: Selective blockade of Endothelin-A (ETA) receptors on glomerular and peritubular cells, inhibiting fibrosis and inflammation without systemic vasodilation.

Mechanism 2: Upregulation of anti-apoptotic pathways in podocytes, preserving glomerular filtration barrier integrity and reducing proteinuria independently of blood pressure reduction.

2. DOSING & ADMINISTRATION

RenalGuard-5 is administered as a once-daily oral tablet (250mg dose). Peak plasma concentration (C_{max}) is achieved in 2.2 hours. Dose adjustment: May be reduced to 125mg in patients with eGFR <30 mL/min/1.73m² or those experiencing hypotension. Bioavailability is enhanced 40% when taken with a light meal.

3. STORAGE & SHELF LIFE

RenalGuard-5 tablets must be stored in sealed containers at 18°C to 25°C, protected from light and moisture. Temperature Warning: Exposure to temperatures exceeding 32°C for more than 36 hours may degrade the active ETA antagonist component, reducing renoprotective potency by up to 45%. Humidity exposure exceeding 60% relative humidity can compromise tablet integrity. Shelf life: 24 months when stored under recommended conditions.